Smart Disaster Response MIS

Complete Normalized Relational Schema · 1NF ➔ 2NF ➔ 3NF ➔ BCNF ➔ 4NF

Legend	■ Primary Key (PK)	◆ *Foreign Key (FK)*	** in schema notation = FK reference*	*PK + FK in composite = owner FK for weak entity*

USER

Group: A — User Management & Authentication

Column Name	Data Type	Key	Constraints	Description
■user_id	INT	**PK**	AUTO_INCREMENT, NOT NULL	Surrogate primary key
username	VARCHAR(50)		UNIQUE, NOT NULL	Unique login username
email	VARCHAR(100)		UNIQUE, NOT NULL	Unique email address
phone	VARCHAR(20)		NULL	Contact phone number
role	ENUM		NOT NULL	admin | emergency_operator | field_officer | warehouse_manager | finance_officer
password_hash	VARCHAR(255)		NOT NULL	bcrypt/argon2 hashed password
is_active	BOOLEAN		DEFAULT TRUE, NOT NULL	Account active flag
created_at	TIMESTAMP		DEFAULT CURRENT_TIMESTAMP	Account creation timestamp

Functional Dependencies:

```
user_id ➔ username, email, phone, role, password_hash,
is_active, created_at
username ➔ user_id, email, phone, role [Candidate Key]
email ➔ user_id, username, phone, role [Candidate Key]
```

Note: Disjoint-total specialization via role ENUM discriminator (Option 8C). Three candidate keys.

NF	Status	Reason
1NF	✓	Atomic values, no repeating groups
2NF	✓	No partial dependencies
3NF	✓	No transitive dependencies
BCNF	✓	Every determinant is a candidate key
4NF	✓	No multi-valued dependencies

CITIZEN

Group: A — User Management & Authentication

Column Name	Data Type	Key	Constraints	Description
■citizen_id	INT	**PK**	AUTO_INCREMENT, NOT NULL	Surrogate primary key
full_name	VARCHAR(100)		NOT NULL	Full legal name
cnic	CHAR(15)		UNIQUE, NULL	Pakistani CNIC (unique when provided)
phone	VARCHAR(20)		NULL	Contact phone
address	TEXT		NULL	Residential address
registered_at	TIMESTAMP		DEFAULT CURRENT_TIMESTAMP	Registration timestamp

Functional Dependencies:

```
citizen_id ➔ full_name, cnic, phone, address, registered_at
cnic ➔ citizen_id, full_name [Candidate Key when NOT NULL]
```

NF	Status	Reason
1NF	✓	Atomic values, no repeating groups
2NF	✓	No partial dependencies
3NF	✓	No transitive dependencies
BCNF	✓	Every determinant is a candidate key
4NF	✓	No multi-valued dependencies

EMERGENCY_REPORT

Group: B — Emergency Reporting

Column Name	Data Type	Key	Constraints	Description
■report_id	INT	**PK**	AUTO_INCREMENT, NOT NULL	Surrogate PK
◆ *citizen_id*	INT	*FK*	NOT NULL, FK➔ CITIZEN	Reporting citizen
◆ *operator_id*	INT	*FK*	NULL, FK➔ USER	Assigned emergency operator
disaster_type	VARCHAR(50)		NOT NULL	flood | earthquake | fire | other
severity_level	ENUM		NOT NULL	low | medium | high | critical

Column Name	Data Type	Key	Constraints	Description
location	VARCHAR(255)		NOT NULL	Textual location description
latitude	DECIMAL(10,7)		NULL	GPS latitude
longitude	DECIMAL(10,7)		NULL	GPS longitude
status	ENUM		NOT NULL, DEFAULT 'pending'	pending | in_progress | resolved | closed
reported_at	TIMESTAMP		DEFAULT CURRENT_TIMESTAMP	Report submission time
resolved_at	TIMESTAMP		NULL	Resolution time (NULL if open)

Functional Dependencies:

`report_id → citizen_id, operator_id, disaster_type, severity_level, location, latitude, longitude, status, reported_at, resolved_at`

NF	Status	Reason
1NF	✓	Atomic values, no repeating groups
2NF	✓	No partial dependencies
3NF	✓	No transitive dependencies
BCNF	✓	Every determinant is a candidate key
4NF	✓	No multi-valued dependencies

MEDIA_ATTACHMENT

Group: B — Emergency Reporting

Column Name	Data Type	Key	Constraints	Description
■ **attachment_id**	INT	**PK**	AUTO_INCREMENT, NOT NULL	Surrogate PK
◆ *report_id*	INT	*FK*	NOT NULL, FK→ EMERGENCY_REPORT	Parent emergency report
file_url	VARCHAR(500)		NOT NULL	Object-storage URL
media_type	VARCHAR(50)		NOT NULL	image | video | audio | document
uploaded_at	TIMESTAMP		DEFAULT CURRENT_TIMESTAMP	Upload timestamp
◆ *uploaded_by*	INT	*FK*	NULL, FK→ USER	Uploader (if authenticated)

Functional Dependencies:

`attachment_id → report_id, file_url, media_type, uploaded_at, uploaded_by`

Note: Weak entity owned by EMERGENCY_REPORT (ER Step 2). Stores file metadata only.

NF	Status	Reason
1NF	✓	Atomic values, no repeating groups
2NF	✓	No partial dependencies
3NF	✓	No transitive dependencies
BCNF	✓	Every determinant is a candidate key
4NF	✓	No multi-valued dependencies

NOTIFICATION

Group: B — Emergency Reporting

Column Name	Data Type	Key	Constraints	Description
■ **notification_id**	INT	**PK**	AUTO_INCREMENT, NOT NULL	Surrogate PK
◆ *user_id*	INT	*FK*	NOT NULL, FK→ USER	Recipient user
message	TEXT		NOT NULL	Notification body
notification_type	VARCHAR(50)		NOT NULL	alert | assignment | approval | system
is_read	BOOLEAN		DEFAULT FALSE, NOT NULL	Read/unread flag
sent_at	TIMESTAMP		DEFAULT CURRENT_TIMESTAMP	Dispatch timestamp

Functional Dependencies:

`notification_id → user_id, message, notification_type, is_read, sent_at`

NF	Status	Reason
1NF	✓	Atomic values, no repeating groups
2NF	✓	No partial dependencies
3NF	✓	No transitive dependencies
BCNF	✓	Every determinant is a candidate key
4NF	✓	No multi-valued dependencies

RESCUE_TEAM

Group: C — Rescue Operations

Column Name	Data Type	Key	Constraints	Description
■ **team_id**	INT	**PK**	AUTO_INCREMENT, NOT NULL	Surrogate PK
team_name	VARCHAR(100)		NOT NULL	Descriptive team name
team_type	ENUM		NOT NULL	medical | fire | rescue
current_location	VARCHAR(255)		NULL	Last known location
availability_status	ENUM		NOT NULL, DEFAULT 'available'	available | assigned | busy | completed
capacity	INT		NOT NULL, DEFAULT 1	Maximum member count

Functional Dependencies:

`team_id → team_name, team_type, current_location, availability_status, capacity`

Note: Status transitions managed by triggers on TEAM_ASSIGNMENT insert/update.

NF	Status	Reason
1NF	✓	Atomic values, no repeating groups
2NF	✓	No partial dependencies
3NF	✓	No transitive dependencies
BCNF	✓	Every determinant is a candidate key
4NF	✓	No multi-valued dependencies

TEAM_MEMBER

Group: C — Rescue Operations

Column Name	Data Type	Key	Constraints	Description
■ **member_id**	INT	**PK**	NOT NULL (partial key)	Partial key within team
■ **team_id**	INT	**PK, FK**	NOT NULL, FK→ RESCUE_TEAM	Owner team — part of composite PK
◆ *user_id*	INT	*FK*	NOT NULL, UNIQUE, FK→ USER	Linked user account
member_role	VARCHAR(50)		NULL	Role within team
joined_at	TIMESTAMP		DEFAULT CURRENT_TIMESTAMP	Date joined team

Functional Dependencies:

`(team_id, member_id) → user_id, member_role, joined_at`

`user_id → (team_id, member_id) [Candidate Key — one team per user]`

Note: Weak entity (ER Step 2). Composite PK ensures full FD — 2NF satisfied. Two CKs → BCNF satisfied.

NF	Status	Reason
1NF	✓	Atomic values, no repeating groups
2NF	✓	No partial dependencies
3NF	✓	No transitive dependencies
BCNF	✓	Every determinant is a candidate key
4NF	✓	No multi-valued dependencies

TEAM_ASSIGNMENT

Group: C — Rescue Operations

Column Name	Data Type	Key	Constraints	Description
■ **assignment_id**	INT	**PK**	AUTO_INCREMENT, NOT NULL	Surrogate PK
◆ *team_id*	INT	*FK*	NOT NULL, FK→ RESCUE_TEAM	Assigned rescue team
◆ *report_id*	INT	*FK*	NOT NULL, FK→ EMERGENCY_REPORT	Target emergency report
assigned_at	TIMESTAMP		DEFAULT CURRENT_TIMESTAMP	Assignment timestamp
completed_at	TIMESTAMP		NULL	Completion time (NULL if ongoing)
status	ENUM		NOT NULL, DEFAULT 'assigned'	assigned | in_progress | completed | cancelled
notes	TEXT		NULL	Field notes

Functional Dependencies:

`assignment_id → team_id, report_id, assigned_at, completed_at, status, notes`

`(team_id, report_id) is an alternate CK if one assignment per pair is enforced.`

Note: M:N junction between RESCUE_TEAM and EMERGENCY_REPORT (ER Step 5).

NF	Status	Reason
1NF	✓	Atomic values, no repeating groups
2NF	✓	No partial dependencies
3NF	✓	No transitive dependencies
BCNF	✓	Every determinant is a candidate key
4NF	✓	No multi-valued dependencies

DISPATCH_LOG

Group: C — Rescue Operations

Column Name	Data Type	Key	Constraints	Description
■ **log_id**	INT	**PK**	NOT NULL (partial key)	Partial key within team
■ **team_id**	INT	**PK, FK**	NOT NULL, FK→ RESCUE_TEAM	Owner team — part of composite PK
status_update	VARCHAR(255)		NOT NULL	Status description at log time
location_update	VARCHAR(255)		NULL	Reported GPS or address
◆ *warehouse_id*	INT	*FK*	NULL, FK→ WAREHOUSE	Warehouse involved (if any)
logged_at	TIMESTAMP		DEFAULT CURRENT_TIMESTAMP	Log entry timestamp

Functional Dependencies:

`(team_id, log_id) → status_update, location_update, warehouse_id, logged_at`

Note: Weak entity (ER Step 2/6). Multi-valued attribute logs extracted from RESCUE_TEAM.

NF	Status	Reason
1NF	✓	Atomic values, no repeating groups
2NF	✓	No partial dependencies
3NF	✓	No transitive dependencies
BCNF	✓	Every determinant is a candidate key
4NF	✓	No multi-valued dependencies

HOSPITAL

Group: D — Hospital & Patient Management

Column Name	Data Type	Key	Constraints	Description
■ **hospital_id**	INT	**PK**	AUTO_INCREMENT, NOT NULL	Surrogate PK
name	VARCHAR(150)		NOT NULL	Hospital name
location	VARCHAR(255)		NOT NULL	Physical address or GPS
total_beds	INT		NOT NULL, DEFAULT 0	Total bed capacity
available_beds	INT		NOT NULL, DEFAULT 0, CHECK >= 0	Current available beds
contact_number	VARCHAR(20)		NULL	Emergency contact number
is_active	BOOLEAN		DEFAULT TRUE, NOT NULL	Operational status flag

Functional Dependencies:

`hospital_id → name, location, total_beds, available_beds, contact_number, is_active`

Note: available_beds decremented/incremented by trigger on PATIENT admission/discharge.

NF	Status	Reason
1NF	✓	Atomic values, no repeating groups
2NF	✓	No partial dependencies
3NF	✓	No transitive dependencies
BCNF	✓	Every determinant is a candidate key
4NF	✓	No multi-valued dependencies

PATIENT

Group: D — Hospital & Patient Management

Column Name	Data Type	Key	Constraints	Description
■ **patient_id**	INT	**PK**	AUTO_INCREMENT, NOT NULL	Surrogate PK
◆ *report_id*	INT	*FK*	NOT NULL, FK→ EMERGENCY_REPORT	Source emergency report
◆ *hospital_id*	INT	*FK*	NULL, FK→ HOSPITAL	Admitted hospital (NULL if unassigned)
◆ *field_officer_id*	INT	*FK*	NULL, FK→ USER	Assigning field officer
admission_time	TIMESTAMP		NOT NULL	Admission timestamp
discharge_time	TIMESTAMP		NULL	Discharge timestamp (NULL if admitted)
condition	VARCHAR(100)		NOT NULL	stable | critical | serious | discharged

Functional Dependencies:

`patient_id → report_id, hospital_id, field_officer_id, admission_time, discharge_time, condition`

NF	Status	Reason
1NF	✓	Atomic values, no repeating groups
2NF	✓	No partial dependencies
3NF	✓	No transitive dependencies
BCNF	✓	Every determinant is a candidate key
4NF	✓	No multi-valued dependencies

WAREHOUSE

Group: E — Warehouse, Resource & Allocation

Column Name	Data Type	Key	Constraints	Description
■**warehouse_id**	INT	**PK**	AUTO_INCREMENT, NOT NULL	Surrogate PK
◆ *manager_id*	INT	*FK*	NULL, FK→ USER	Assigned warehouse manager
name	VARCHAR(150)		NOT NULL	Warehouse identifier
location	VARCHAR(255)		NOT NULL	Physical address
total_capacity	INT		NOT NULL, DEFAULT 0	Maximum storage capacity (units)
created_at	TIMESTAMP		DEFAULT CURRENT_TIMESTAMP	Record creation timestamp

Functional Dependencies:

`warehouse_id → manager_id, name, location, total_capacity, created_at`

NF	Status	Reason
1NF	✓	Atomic values, no repeating groups
2NF	✓	No partial dependencies
3NF	✓	No transitive dependencies
BCNF	✓	Every determinant is a candidate key
4NF	✓	No multi-valued dependencies

RESOURCE

Group: E — Warehouse, Resource & Allocation

Column Name	Data Type	Key	Constraints	Description
■**resource_id**	INT	**PK**	AUTO_INCREMENT, NOT NULL	Surrogate PK
resource_name	VARCHAR(100)		NOT NULL	E.g. Rice 50kg bag
resource_type	VARCHAR(50)		NOT NULL	food | water | medicine | shelter | equipment
unit_of_measure	VARCHAR(20)		NOT NULL	kg | litre | unit | box
description	TEXT		NULL	Additional description

Functional Dependencies:

`resource_id → resource_name, resource_type, unit_of_measure, description`

Note: Catalogue of resource types. Actual per-warehouse quantities stored in WAREHOUSE_INVENTORY.

NF	Status	Reason
1NF	✓	Atomic values, no repeating groups
2NF	✓	No partial dependencies
3NF	✓	No transitive dependencies
BCNF	✓	Every determinant is a candidate key
4NF	✓	No multi-valued dependencies

WAREHOUSE_INVENTORY

Group: E — Warehouse, Resource & Allocation

Column Name	Data Type	Key	Constraints	Description
■**warehouse_id**	INT	**PK, FK**	NOT NULL, FK→ WAREHOUSE	Owner warehouse — part of composite PK
■**resource_id**	INT	**PK, FK**	NOT NULL, FK→ RESOURCE	Stocked resource — part of composite PK
quantity_available	DECIMAL(12,2)		NOT NULL, DEFAULT 0, CHECK >= 0	Current stock quantity
threshold_level	DECIMAL(12,2)		NOT NULL, DEFAULT 0	Low-stock alert threshold
last_updated	TIMESTAMP		DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Last modification timestamp

Functional Dependencies:

`(warehouse_id, resource_id) → quantity_available, threshold_level, last_updated`

`warehouse_id alone ■ quantity_available | resource_id alone ■ quantity_available`

Note: Weak entity with dual ownership. Composite PK is the only CK — BCNF trivially satisfied.

NF	Status	Reason
1NF	✓	Atomic values, no repeating groups
2NF	✓	No partial dependencies
3NF	✓	No transitive dependencies
BCNF	✓	Every determinant is a candidate key
4NF	✓	No multi-valued dependencies

RESOURCE_ALLOCATION

Group: E — Warehouse, Resource & Allocation

Column Name	Data Type	Key	Constraints	Description
■ **allocation_id**	INT	**PK**	AUTO_INCREMENT, NOT NULL	Surrogate PK
◆ *report_id*	INT	*FK*	NOT NULL, FK→ EMERGENCY_REPORT	Requesting emergency
◆ *resource_id*	INT	*FK*	NOT NULL, FK→ RESOURCE	Resource being allocated
◆ *warehouse_id*	INT	*FK*	NOT NULL, FK→ WAREHOUSE	Source warehouse
qty_requested	DECIMAL(12,2)		NOT NULL, CHECK > 0	Requested quantity
qty_dispatched	DECIMAL(12,2)		NOT NULL, DEFAULT 0, CHECK >= 0	Quantity physically dispatched
qty_consumed	DECIMAL(12,2)		NOT NULL, DEFAULT 0, CHECK >= 0	Quantity confirmed consumed
status	ENUM		NOT NULL, DEFAULT 'pending'	pending | approved | dispatched | consumed | rejected
requested_at	TIMESTAMP		DEFAULT CURRENT_TIMESTAMP	Request submission time
◆ *approved_by*	INT	*FK*	NULL, FK→ USER	Approving user

Functional Dependencies:

`allocation_id → report_id, resource_id, warehouse_id, qty_requested, qty_dispatched, qty_consumed, status, requested_at, approved_by`

Note: Trigger: On status→ dispatched, deducts qty_dispatched from WAREHOUSE_INVENTORY.quantity_available.

NF	Status	Reason
1NF	✓	Atomic values, no repeating groups
2NF	✓	No partial dependencies
3NF	✓	No transitive dependencies
BCNF	✓	Every determinant is a candidate key
4NF	✓	No multi-valued dependencies

APPROVAL_REQUEST

Group: E — Warehouse, Resource & Allocation

Column Name	Data Type	Key	Constraints	Description
■ **approval_id**	INT	**PK**	AUTO_INCREMENT, NOT NULL	Surrogate PK
◆ *allocation_id*	INT	*FK*	NULL, FK→ RESOURCE_ALLOCATION	Related allocation (if applicable)
request_type	VARCHAR(60)		NOT NULL	resource_allocation | financial | deployment
reference_id	INT		NULL	PK of entity being approved (polymorphic)
status	ENUM		NOT NULL, DEFAULT 'pending'	pending | approved | rejected
requested_at	TIMESTAMP		DEFAULT CURRENT_TIMESTAMP	Request creation timestamp
reviewed_at	TIMESTAMP		NULL	Review timestamp (NULL if pending)
remarks	TEXT		NULL	Reviewer comments
◆ *requested_by*	INT	*FK*	NOT NULL, FK→ USER	Submitting user
◆ *reviewed_by*	INT	*FK*	NULL, FK→ USER	Reviewing user (NULL if pending)

Functional Dependencies:

`approval_id → allocation_id, request_type, reference_id, status, requested_at, reviewed_at, remarks, requested_by, reviewed_by`

NF	Status	Reason
1NF	✓	Atomic values, no repeating groups
2NF	✓	No partial dependencies
3NF	✓	No transitive dependencies
BCNF	✓	Every determinant is a candidate key
4NF	✓	No multi-valued dependencies

DONATION

Group: F — Financial Management

Column Name	Data Type	Key	Constraints	Description
■ **donation_id**	INT	**PK**	AUTO_INCREMENT, NOT NULL	Surrogate PK
◆ *received_by*	INT	*FK*	NULL, FK→ USER	Finance officer who received
donor_name	VARCHAR(150)		NOT NULL	Individual or organization name
donor_type	ENUM		NOT NULL	individual | organization | government

Column Name	Data Type	Key	Constraints	Description
amount	DECIMAL(15,2)		NOT NULL, CHECK > 0	Donated amount (PKR)
payment_method	VARCHAR(50)		NULL	cash | bank_transfer | online | cheque
donated_at	TIMESTAMP		DEFAULT CURRENT_TIMESTAMP	Donation timestamp

Functional Dependencies:

`donation_id → received_by, donor_name, donor_type, amount, payment_method, donated_at`

NF	Status	Reason
1NF	✓	Atomic values, no repeating groups
2NF	✓	No partial dependencies
3NF	✓	No transitive dependencies
BCNF	✓	Every determinant is a candidate key
4NF	✓	No multi-valued dependencies

EXPENSE

Group: F — Financial Management

Column Name	Data Type	Key	Constraints	Description
■ **expense_id**	INT	**PK**	AUTO_INCREMENT, NOT NULL	Surrogate PK
◆ *recorded_by*	INT	*FK*	NOT NULL, FK→ USER	Finance officer who recorded
◆ *allocation_id*	INT	*FK*	NULL, FK→ RESOURCE_ALLOCATION	Related allocation (if procurement)
category	VARCHAR(100)		NOT NULL	procurement | logistics | medical | admin | other
amount	DECIMAL(15,2)		NOT NULL, CHECK > 0	Expense amount (PKR)
description	TEXT		NULL	Detailed description
expense_date	DATE		NOT NULL	Date of expense

Functional Dependencies:

`expense_id → recorded_by, allocation_id, category, amount, description, expense_date`

NF	Status	Reason
1NF	✓	Atomic values, no repeating groups
2NF	✓	No partial dependencies
3NF	✓	No transitive dependencies
BCNF	✓	Every determinant is a candidate key
4NF	✓	No multi-valued dependencies

FINANCE_TRANSACTION

Group: F — Financial Management

Column Name	Data Type	Key	Constraints	Description
■ **transaction_id**	INT	**PK**	AUTO_INCREMENT, NOT NULL	Surrogate PK
◆ *performed_by*	INT	*FK*	NOT NULL, FK→ USER	Finance officer performing
◆ *donation_id*	INT	*FK*	NULL, FK→ DONATION	Linked donation (NULL for expenses)
◆ *expense_id*	INT	*FK*	NULL, FK→ EXPENSE	Linked expense (NULL for donations)
transaction_type	ENUM		NOT NULL	donation | expense | procurement | transfer
amount	DECIMAL(15,2)		NOT NULL, CHECK > 0	Transaction amount (PKR)
transaction_timestamp	TIMESTAMP		DEFAULT CURRENT_TIMESTAMP	Transaction timestamp
status	ENUM		NOT NULL, DEFAULT 'completed'	pending | completed | failed | reversed

Functional Dependencies:

`transaction_id → performed_by, donation_id, expense_id, transaction_type,`
`amount, transaction_timestamp, status`

Note: XOR constraint: (donation_id IS NULL) XOR (expense_id IS NULL). Enforced via CHECK or trigger.

NF	Status	Reason
1NF	✓	Atomic values, no repeating groups
2NF	✓	No partial dependencies
3NF	✓	No transitive dependencies
BCNF	✓	Every determinant is a candidate key
4NF	✓	No multi-valued dependencies

AUDIT_LOG

Group: G — Audit & Monitoring

Column Name	Data Type	Key	Constraints	Description
■ **log_id**	BIGINT	**PK**	AUTO_INCREMENT, NOT NULL	Surrogate PK (BIGINT for volume)
◆ *user_id*	INT	*FK*	NULL, FK→ USER	Acting user (NULL for system events)
action_type	VARCHAR(20)		NOT NULL	INSERT | UPDATE | DELETE | LOGIN | LOGOUT
table_affected	VARCHAR(60)		NOT NULL	Name of the affected table
record_id	BIGINT		NULL	PK of the affected row
old_value	JSON		NULL	Before-image (NULL for INSERTs)
new_value	JSON		NULL	After-image (NULL for DELETEs)
ip_address	VARCHAR(45)		NULL	Client IPv4 or IPv6
action_timestamp	TIMESTAMP		NOT NULL, DEFAULT CURRENT_TIMESTAMP	Exact action timestamp

Functional Dependencies:

`log_id → user_id, action_type, table_affected, record_id, old_value, new_value, ip_address, action_timestamp`

Note: IMMUTABLE: REVOKE UPDATE, DELETE ON AUDIT_LOG FROM all roles. INSERT only via triggers/middleware.

NF	Status	Reason
1NF	✓	Atomic values, no repeating groups
2NF	✓	No partial dependencies
3NF	✓	No transitive dependencies
BCNF	✓	Every determinant is a candidate key
4NF	✓	No multi-valued dependencies

Relational Schema — Compact Notation

Underlined = Primary Key · Asterisk (*) = Foreign Key · + = composite PK

Notation Guide	PK_col → Primary Key attribute (underlined in academic notation)	attr* → Foreign Key reference to another table	col1 + col2 → Composite Primary Key (weak entity)

A — User Management & Authentication

USER (**user_id**, username, email, phone, role, password_hash, is_active, created_at)

CITIZEN (**citizen_id**, full_name, cnic, phone, address, registered_at)

B — Emergency Reporting

EMERGENCY_REPORT (**report_id**, *citizen_id**, *operator_id**, disaster_type, severity_level, location, latitude, longitude, status, reported_at, resolved_at)

MEDIA_ATTACHMENT (**attachment_id**, *report_id**, *uploaded_by**, file_url, media_type, uploaded_at)

NOTIFICATION (**notification_id**, *user_id**, message, notification_type, is_read, sent_at)

C — Rescue Operations

RESCUE_TEAM (**team_id**, team_name, team_type, current_location, availability_status, capacity)

TEAM_MEMBER (**team_id**, **member_id**, *user_id**, member_role, joined_at)

TEAM_ASSIGNMENT (**assignment_id**, *team_id**, *report_id**, assigned_at, completed_at, status, notes)

DISPATCH_LOG (**team_id**, **log_id**, status_update, location_update, *warehouse_id**, logged_at)

D — Hospital & Patient Management

HOSPITAL (**hospital_id**, name, location, total_beds, available_beds, contact_number, is_active)

PATIENT (**patient_id**, *report_id**, *hospital_id**, *field_officer_id**, admission_time, discharge_time, condition)

E — Warehouse, Resource & Allocation

WAREHOUSE (**warehouse_id**, *manager_id**, name, location, total_capacity, created_at)

RESOURCE (**resource_id**, resource_name, resource_type, unit_of_measure, description)

WAREHOUSE_INVENTORY (**warehouse_id**, **resource_id**, quantity_available, threshold_level, last_updated)

RESOURCE_ALLOCATION (**allocation_id**, *report_id**, *resource_id**, *warehouse_id**, qty_requested, qty_dispatched, qty_consumed, status, requested_at, *approved_by**)

APPROVAL_REQUEST (**approval_id**, *allocation_id**, request_type, reference_id, status, requested_at, reviewed_at, remarks, *requested_by**, *reviewed_by**)

F — Financial Management

DONATION (**donation_id**, *received_by**, donor_name, donor_type, amount, payment_method, donated_at)

EXPENSE (**expense_id**, *recorded_by**, *allocation_id**, category, amount, description, expense_date)

FINANCE_TRANSACTION (**transaction_id**, *performed_by**, *donation_id**, *expense_id**, transaction_type, amount, transaction_timestamp, status)

G — Audit & Monitoring

AUDIT_LOG (**log_id**, *user_id**, action_type, table_affected, record_id, old_value, new_value, ip_address, action_timestamp)

Relational Schema — Visual Box Notation

Each table shown as a named box with attribute cells · PK in bold red · FK in italic blue

A — User Management & Authentication

USER	**user_id**	username	email	phone	role	password_hash	is_active	created_at

CITIZEN	**citizen_id**	full_name	cnic	phone	address	registered_at

B — Emergency Reporting

EMERGENCY_REPORT	**report_id**	*citizen_id*	*operator_id*	disaster_type	severity_level	location	latitude	longitude	status	reported_at	resolved_at

MEDIA_ATTACHMENT	**attachment_id**	*report_id*	file_url	media_type	uploaded_at	*uploaded_by*

NOTIFICATION	**notification_id**	*user_id*	message	notification_type	is_read	sent_at

C — Rescue Operations

RESCUE_TEAM	**team_id**	team_name	team_type	current_location	availability_status	capacity

TEAM_MEMBER	**team_id**	**member_id**	*user_id*	member_role	joined_at

TEAM_ASSIGNMENT	**assignment_id**	*team_id*	*report_id*	assigned_at	completed_at	status	notes

DISPATCH_LOG	**team_id**	**log_id**	status_update	location_update	*warehouse_id*	logged_at

D — Hospital & Patient Management

HOSPITAL	**hospital_id**	name	location	total_beds	available_beds	contact_number	is_active

PATIENT	**patient_id**	*report_id*	*hospital_id*	*field_officer_id*	admission_time	discharge_time	condition

E — Warehouse, Resource & Allocation

WAREHOUSE	**warehouse_id**	*manager_id*	name	location	total_capacity	created_at

RESOURCE	**resource_id**	resource_name	resource_type	unit_of_measure	description

WAREHOUSE_INVENTORY	**warehouse_id**	**resource_id**	quantity_available	threshold_level	last_updated

RESOURCE_ALLOCATION	**allocation_id**	*report_id*	*resource_id*	*warehouse_id*	qty_requested	qty_dispatched	qty_consumed	status	requested_at	*approved_by*

APPROVAL_REQUEST	**approval_id**	*allocation_id*	request_type	reference_id	status	requested_at	reviewed_at	remarks	*requested_by*	*reviewed_by*

F — Financial Management

DONATION	**donation_id**	*received_by*	donor_name	donor_type	amount	payment_method	donated_at

EXPENSE	**expense_id**	*recorded_by*	*allocation_id*	category	amount	description	expense_date

FINANCE_TRANSACTION	**transaction_id**	*performed_by*	*donation_id*	*expense_id*	transaction_type	amount	transaction_timestamp	status

G — Audit & Monitoring

AUDIT_LOG	**log_id**	*user_id*	action_type	table_affected	record_id	old_value	new_value	ip_address	action_timestamp